LAB - 6

Huffman Coding & Job Sequencing Problem

Huffman Coding:

CODE:

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

struct Node {
   char data;
   unsigned freq;
   struct Node *left, *right;
};

struct Node* newNode(char data, unsigned freq) {
   struct Node* temp = (struct
Node*)malloc(sizeof(struct Node));
   temp->left = temp->right = NULL;
   temp->data = data;
   temp->freq = freq;
   return temp;
}
struct MinHeap {
   unsigned size;
   struct Node* arr[MAX];
};
void swap(struct Node** a, struct Node** b) {
   struct Node* t = *a;
   *a = *b;
   *b = t;
}
void heapify(struct MinHeap* heap, int i) {
   int smallest = i;
   int left = 2 * i + 1;
   int right = 2 * i + 2;
   if (left < heap->size && heap->arr[left]->freq <
heap->arr[smallest]->freq)
      smallest = left;
   if (right < heap->size && heap->arr[right]->freq
< heap->arr[smallest]->freq)
```

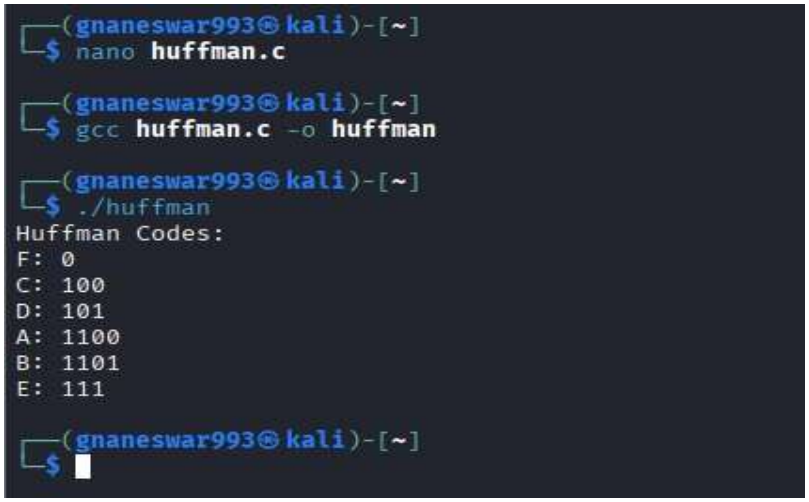
```
        smallest = right;
    if (smallest != i) {
        swap(&heap->arr[i], &heap->arr[smallest]);
        heapify(heap, smallest);
    }
}
struct Node* extractMin(struct MinHeap* heap) {
    struct Node* temp = heap->arr[0];
    heap->arr[0] = heap->arr[--heap->size];
    heapify(heap, 0);
    return temp;
}
void insertHeap(struct MinHeap* heap, struct Node*
node) {
    int i = heap->size++;

    while (i && node->freq < heap->arr[(i - 1) / 2]-
>freq) {
        heap->arr[i] = heap->arr[(i - 1) / 2];
        i = (i - 1) / 2;
    }
    heap->arr[i] = node;
}
struct Node* buildTree(char data[], int freq[], int size)
{
    struct MinHeap heap;
    heap.size = 0;
    for (int i = 0; i < size; i++)
        heap.arr[heap.size++] = newNode(data[i],
freq[i]);
    for (int i = (heap.size - 1) / 2; i >= 0; i--)
        heapify(&heap, i);
    while (heap.size > 1) {
        struct Node* left = extractMin(&heap);
        struct Node* right = extractMin(&heap);
        struct Node* top = newNode('$', left->freq +
right->freq);
        top->left = left;
        top->right = right;
        insertHeap(&heap, top);
    }
    return extractMin(&heap);
}
```

```
void printCodes(struct Node* root, int arr[], int top)
{
    if (root->left) {
        arr[top] = 0;
        printCodes(root->left, arr, top + 1);
    }
    if (root->right) {
        arr[top] = 1;
        printCodes(root->right, arr, top + 1);
    }
    if (!root->left && !root->right) {
        printf("%c: ", root->data);
        for (int i = 0; i < top; i++)
            printf("%d", arr[i]);
        printf("\n");
    }
}
int main() {
    char data[] = {'A', 'B', 'C', 'D', 'E', 'F'};
    int freq[] = {5, 9, 12, 13, 16, 45};
    int size = sizeof(data) / sizeof(data[0]);
    struct Node* root = buildTree(data, freq, size);

    int arr[MAX], top = 0;
    printf("Huffman Codes:\n");
    printCodes(root, arr, top);
    return 0;
}
```

OUTPUT:



Job Sequencing Problem

CODE:

```
 #include <stdio.h>

#include <stdlib.h>

typedef struct {

    int id;

    int deadline;

    int profit;

} Job;

int compare(const void *a, const void *b) {

    Job *j1 = (Job *)a;

    Job *j2 = (Job *)b;

    return j2->profit - j1->profit;

}

int main() {

    int n;

    printf("Enter number of jobs: ");

    scanf("%d", &n);

    Job jobs[n];


    for (int i = 0; i < n; i++) {

        printf("Enter deadline and profit for job %d: ", i
+ 1);

        scanf("%d %d", &jobs[i].deadline,
&jobs[i].profit);

        jobs[i].id = i + 1;

    }


    qsort(jobs, n, sizeof(Job), compare);

    int maxDeadline = 0;
```

```
    for (int i = 0; i < n; i++) {
        if (jobs[i].deadline > maxDeadline)
            maxDeadline = jobs[i].deadline;
    }
    int slot[maxDeadline];
    for (int i = 0; i < maxDeadline; i++)
        slot[i] = -1;
    int totalProfit = 0;
    for (int i = 0; i < n; i++) {
        for (int j = jobs[i].deadline - 1; j >= 0; j--) {
            if (slot[j] == -1) {
                slot[j] = jobs[i].id;
                totalProfit += jobs[i].profit;
                break;
            }
        }
    }
    printf("\nScheduled Jobs: ");
    for (int i = 0; i < maxDeadline; i++) {
        if (slot[i] != -1)
            printf("J%d ", slot[i]);
    }
    printf("\nTotal Profit = %d\n", totalProfit);
    return 0;
}
```

OUTPUT:

```
┌──(gnaneswar993㉿kali)-[~]
└─$ nano job_sequencing.c

┌──(gnaneswar993㉿kali)-[~]
└─$ gcc job_sequencing.c -o job_sequencing

┌──(gnaneswar993㉿kali)-[~]
└─$ ./job_sequencing
Enter number of jobs: 4
Enter deadline and profit for job 1: 4 20
Enter deadline and profit for job 2: 1 10
Enter deadline and profit for job 3: 1 40
Enter deadline and profit for job 4: 1 30

Scheduled Jobs: J3 J1
Total Profit = 60

┌──(gnaneswar993㉿kali)-[~]
└─$ 
```

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